COMPUTER ENGINEERING
QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE010 | **SECTION:** | CPE21S2 |
| **DATE PERFORMED:** | 7/15/26 | **DATE SUBMITTED:** | 7/15/26 |
| **NAME:** | Gabriel Dwyane D. Aridon | **INSTRUCTOR:** | Engr. Jimlord Quejado |

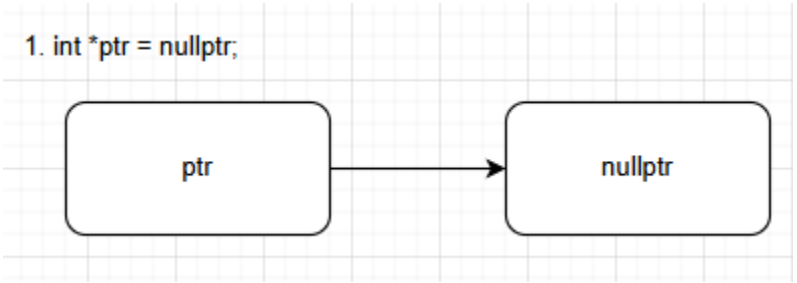
## ACTIVITY NO. 2.1 Memory Diagram Exercise

### 1. PROCEDURE OUTPUT

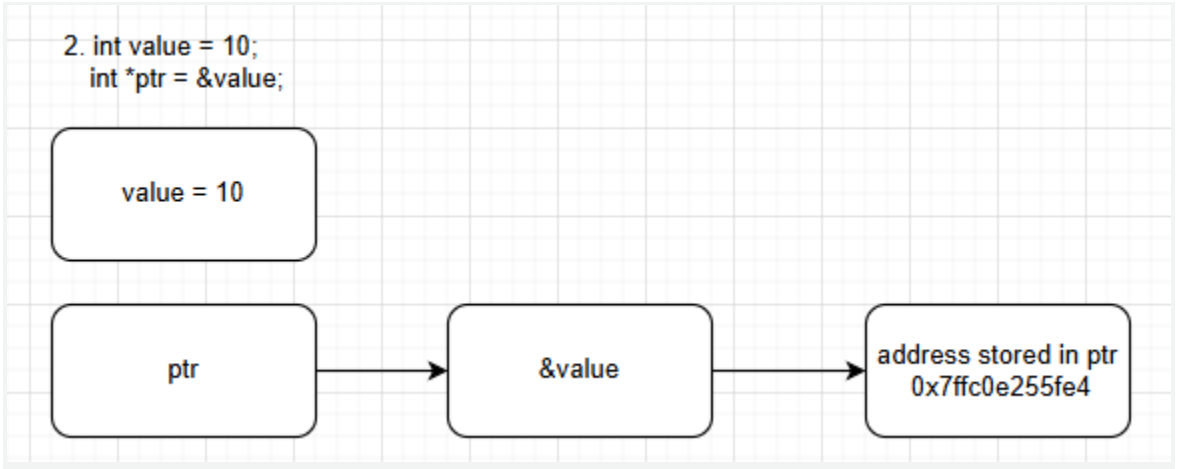
What procedures did you perform?

Draw the memory diagram for the following:

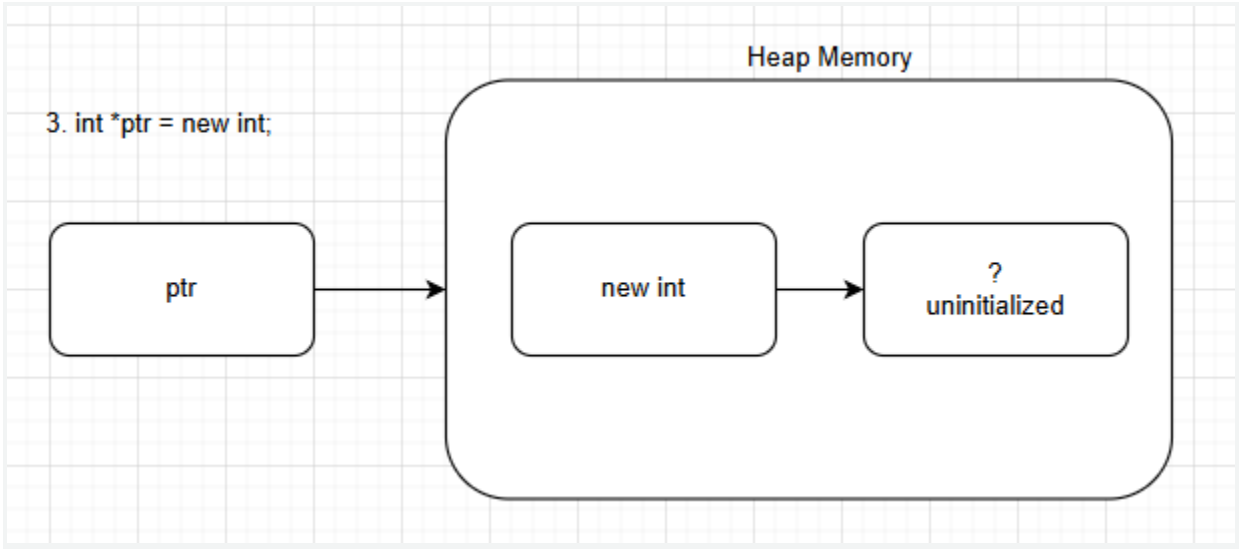
1. int *ptr = nullptr;



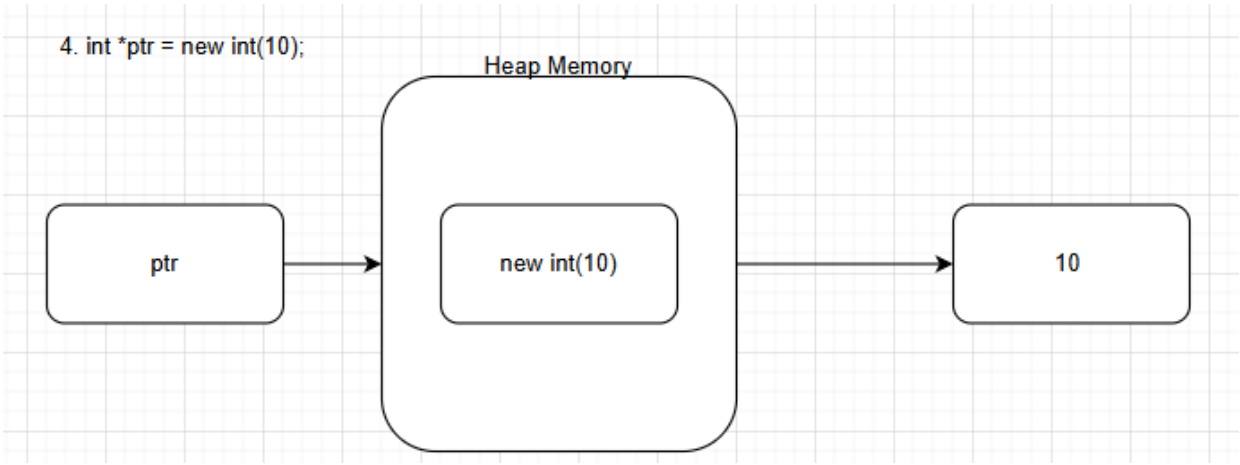

2. int value = 10;
int *ptr= &value;




3. int *ptr = new int;




4. int *ptr = new int(10);




5. Differentiate Constructor, Destructor, Copy Constructor, Copy Assignment Operator.

- A destructor is another special member function that is automatically called when an object is destroyed or leaves the scope because its function is to clean up resources, whereas a constructor is a special method that is automatically called when an object or class is created. When a new object is initialized using another object of the same class, a copy constructor is called to create a new object by copying an existing object. A copy assignment operator, on the other hand, copies the contents of an already-created object to another existing object. In contrast to the copy constructor, it replaces the data of the existing object rather than creating a new one.

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## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **Gemini, Quillbot**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **I used it for explaining the concept and paraphrasing**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **I prompt it to explain to me how it works since it is confusing for me. I also asked if my drawing is correct and what certain adjustments I can do to make the logic correct and accurate.**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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